SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I ____B.Theerthi(192473026)_________________ (Name / Reg No) certify that this submission is my original work and that I have adhered
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

B.Theerthi

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Research Article Review

Article Title

An Artificial Intelligence Approach to Monitor Student Performance and Devise Preventive Measures

Article Citation Details

Authors: Ijaz Khan, Abdul Rahim Ahmad, Nafaa Jabeur, and Mohammed Najah Mahdi
Journal: *Smart Learning Environments*
Volume: 8
Article Number: 17
Year: 2021
DOI: 10.1186/s40561-021-00161-y
Publisher: Springer Nature

DOI Link: https://doi.org/10.1186/s40561-021-00161-y

The article was published on 8 September 2021 and is available as an open-access research article through Springer.

Task 1: Article Summary

(a) Domain/Application Area and ML Problem

The article belongs to the Educational Data Mining and Learning Analytics domain. The application area is student academic performance prediction.

The main machine learning problem addressed is the early identification of students who are likely to obtain an unsatisfactory final result. The researchers wanted to predict whether a student would have a "Low" or "High" final academic outcome before the end of the semester. This would allow instructors to identify struggling students early and provide additional support.

The study evaluates several supervised machine learning techniques, including Decision Tree, k-Nearest Neighbours (k-NN), Artificial Neural Network (ANN), and Naïve Bayes. Among these, the Decision Tree model performed best overall and was selected because its rules could be easily understood by instructors.

(b) Key Objective and Research Gap

The main objective of the research was to develop a machine learning-based system that could identify academically struggling students at an early stage of a course. Educational institutions collect large amounts of student information, but simply collecting data does not automatically help instructors identify students who need support. The authors observed that existing student-performance prediction models were often developed for specific educational environments and were not always transformed into understandable recommendations that instructors could directly use. The research therefore aimed not only to build a prediction model but also to convert its results into understandable decision rules and preventive actions. The authors prepared student academic data, applied several supervised learning algorithms, compared their performance, and selected the best-performing model. The Decision Tree achieved the strongest overall performance and was converted into interpretable rules based mainly on CGPA, examination performance, and attendance. The research therefore attempted to bridge the gap between prediction and practical educational intervention, allowing instructors to identify at-risk students early and provide personalized support.

Task 2: Methodology Analysis

(a) ML Techniques Used

The study uses supervised machine learning classification techniques.

The four main algorithms evaluated were:

1. **Decision Tree – RepTree**
2. **k-Nearest Neighbours (k-NN)**
3. **Artificial Neural Network – Multilayer Perceptron**
4. **Naïve Bayes**

The final selected technique was the Decision Tree.

A Decision Tree is an inductive learning technique. It learns rules from previously labelled examples and uses these rules to classify new instances. The tree recursively divides the data based on important features. In this study, the resulting tree was easy for instructors to interpret.

Dataset

The dataset was obtained from Buraimi University College (BUC), Sultanate of Oman. It contained student academic records from a face-to-face course called Phonetics and Phonology.

The original training dataset contained:

- 151 student instances

- 10 prediction features
- 1 prediction class
- Target classes: Low / High
- Data collected over three semesters

The features included academic and attendance-related information. After feature selection, the researchers selected four important features:

- CGPA
- Pre-requisite course grades
- Exam-1 marks
- Attendance

The researchers removed irrelevant information such as student ID, student name and course code because of privacy concerns. They also removed noisy or incomplete instances. Gain Ratio Attribute Evaluator with Ranker was then used for feature selection.

Experimental Procedure

The experiments were performed using WEKA. The researchers used 10-fold cross-validation for model evaluation. In this procedure, the dataset is divided into ten parts. Nine parts are used for training and one part for testing, with the testing portion changed during each iteration.

(b) Mapping to CO 4

CO 4 Topic	Connection with Article
Inductive Learning	The model learns classification rules from previously observed student records and applies them to unseen students.
Decision Trees	RepTree is the main technique and produces an interpretable tree structure.
Statistical Learning	Naïve Bayes is used as one of the benchmark classifiers.
Reinforcement Learning	Not directly used in this research.

Methodological Strength

The major strength is **interpretability**. Instead of producing only a prediction, the Decision Tree provides understandable rules. For example, the study found that CGPA, exam-1 performance and attendance were important factors in predicting student performance. This makes the model practical for teachers because they can understand why a student is classified as at risk.

Methodological Limitation

A major limitation is the small and institution-specific dataset. Only 151 training instances were used, and the data came from one course at one university. Therefore, the model may not generalize well to students from different universities, courses, countries, or educational systems. The authors themselves describe the existing prediction models as being strongly dependent on their local educational context.

Task 3: Results and Critical Evaluation

(a) Key Results

The researchers compared the four machine learning algorithms using accuracy, precision, recall, F-Measure and Matthews Correlation Coefficient (MCC).

The Decision Tree performed best overall.

Model	Overall Result
Decision Tree (RepTree)	Best overall performance
ANN / MLP	Competitive, generally second-best
k-NN	Good recall but lower overall performance
Naïve Bayes	Good precision but weaker overall performance

The authors report that the Decision Tree achieved:

- **Accuracy: over 86%**
- **F-Measure: 0.91**
- **MCC: 0.63**

The paper also reports that the Decision Tree achieved the highest F-Measure and MCC among the evaluated models. The Decision Tree had relatively lower recall than some other models, although it achieved the highest accuracy.

The authors therefore selected the Decision Tree for further interpretation and implementation.

Important Decision Tree Findings

The resulting tree showed that:

- CGPA was an important predictor.
- Exam-1 marks were another important factor.
- Attendance was also significant.
- Students with lower CGPA, lower exam performance and lower attendance were more likely to be classified as having a low final outcome.

(b) Critical Evaluation

The reported results are reasonably convincing for the **specific dataset**, but they should not be interpreted as proof that the model will achieve the same performance in other institutions.

Are the metrics realistic?

An accuracy above 86% and F-Measure of 0.91 are possible for a relatively structured educational dataset. However, the dataset contains only 151 instances, which is small for machine learning research. Therefore, the reported performance could be affected by the characteristics of this particular dataset.

The researchers correctly used several metrics instead of depending only on accuracy. This is important because the classes may not be evenly distributed. MCC is particularly useful because it considers all four components of the confusion matrix.

Potential Biases

There are several possible sources of bias:

1.Datasetbias:
The data comes from one university and one course. Student behaviour in another institution may be different.

2.Smallsamplesize:
151 instances can make performance estimates less stable.

3.Classimbalance:
The authors themselves discuss the problem of unequal class distribution in student-performance datasets. Accuracy alone may therefore be misleading.

4.Featurebias:
CGPA, attendance and exam performance are useful indicators, but factors such as socioeconomic background, learning behaviour, access to technology, motivation and study hours may also influence performance.

Additional Experiments That Would Strengthen the Research

The research could be strengthened by:

- Testing the model on datasets from multiple universities.
- Increasing the number of students and courses.
- Performing external validation on a completely independent dataset.
- Comparing the Decision Tree with Random Forest, XGBoost and SVM.
- Reporting confidence intervals for performance metrics.

- Performing repeated cross-validation.
- Testing the model across different semesters and academic years.
- Evaluating fairness across different demographic groups.
- Measuring whether interventions actually improve student outcomes compared with a control group.

These experiments would provide stronger evidence that the model generalizes beyond the original institution.

(c) Real-World Applicability

The proposed system can be deployed in:

- Universities and colleges
- Learning Management Systems (LMS)
- Online education platforms
- Academic advising systems
- Student early-warning systems
- Intelligent tutoring systems

An instructor could run the model after an early examination and identify students who are likely to struggle. The instructor could then provide additional classes, academic counselling or personalized recommendations. The authors actually tested their model in a real classroom with 25 students and identified five students at risk. Four of those five students eventually achieved acceptable final results after support was provided.

Challenges in Industry/Institutional Scaling

Data availability: Different institutions may store student information in different formats.

Data quality: Missing values, inconsistent records and noisy data can reduce prediction quality.

Privacy: Student academic records are sensitive, so institutions must protect personal information.

Generalization: A model trained at one university may not work equally well at another.

Ethical issues: Predictions should be used to support students rather than label them permanently as “weak” students.

Scalability: Large universities may have thousands of students, courses and historical records, requiring automated data pipelines and reliable computing infrastructure.

Task 4: Reflection and Learning Outcome

(a) Three Key Insights

Insight 1: Decision Trees provide both prediction and explanation

The first important insight is that a Decision Tree does not only classify data; it can also provide understandable rules.

Connection to CO 4: This directly relates to the Decision Tree topic. The tree recursively divides the dataset according to important attributes and produces decision rules.

This helped me understand why Decision Trees are useful when interpretability is important.

Insight 2: Accuracy alone is not enough

The article uses accuracy, precision, recall, F-Measure and MCC instead of relying only on accuracy.

Connection to CO 4: This connects to Statistical Learning and model evaluation.

I learned that a model with high accuracy may still perform poorly on one class, particularly when the dataset is imbalanced. F-Measure and MCC provide additional information about classification quality.

Insight 3: Inductive learning depends strongly on training data

The Decision Tree learns patterns from historical student records and applies them to new students.

Connection to CO 4: This connects directly to **Inductive Learning**.

The model's predictions are based on patterns learned from examples. Therefore, if the training dataset is small or biased, the learned rules may not generalize to a different population.

This helped me understand the importance of representative datasets in machine learning.

(b) Proposed Extension

If I were to extend this research, I would develop a multi-university student performance prediction system using an ensemble of Decision Trees such as Random Forest and XGBoost.

The new system could combine academic information such as CGPA and examination marks with attendance, study hours, learning-platform activity and assignment submission behaviour. The model could be trained using datasets collected from multiple universities rather than a single institution.

The experiment could compare the original RepTree with Decision Tree, Random Forest, XGBoost and SVM using repeated cross-validation and an independent test dataset. Accuracy, precision, recall, F1-score, MCC and AUC-ROC could be reported.

This extension is feasible because universities and LMS platforms already generate large amounts of student activity data. The expected impact would be better generalization and more reliable early identification of students who require academic support. An explainability method such as SHAP could also be used so that instructors understand which features influenced each prediction.

Overall Critical Review

The article provides a practical application of Decision Tree-based inductive learning in education. Its major contribution is not simply predicting student performance but converting machine learning predictions into understandable rules and preventive actions. The Decision Tree achieved over 86% accuracy, an F-Measure of 0.91 and an MCC of 0.63, outperforming the other evaluated approaches overall.

The strongest aspect of the study is its interpretability and real-world implementation. However, the relatively small dataset and single-institution setting limit the generalizability of the findings. Larger multi-institution datasets and independent external validation would make the conclusions more reliable.

Overall, this article is highly relevant to CO 4, particularly Inductive Learning and Decision Trees, because it demonstrates how a theoretical machine learning technique can be applied to a real-world problem and converted into actionable decisions for educators.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand